Existence of a Unique Invariant Measure and Ergodic Property in AIMD-based Multi-resource Allocation

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***Abstract*— Distributed resource allocation arises in many application domains, such as smart energy systems, intelligent transportation systems, cloud computing, edge computing, etcetera. To realize many of these applications, agents in a network may require multiple shared resources to complete a task and aim to maximize the network utility. Additionally, they may demand resources based on their preferences. Furthermore, they may not wish to share their cost functions, partial derivatives of the cost functions, etc., with other agents or a central server; however, they share their resource demands with the central server that aggregates the demands and sends one-bit resource-capacity constraint notification in the network. The single-resource allocation algorithms are inefficient and provide sub-optimal solutions for multi-resource allocations, especially when the cost functions are multi-variate and non-separable. We present additive increase and multiplicative decrease algorithm (AIMD)-based distributed solutions for multi-resource allocation. We formulate the resource allocations problem over finite window sizes and model the system as a homogeneous Markov chain with place-dependent probabilities. We show that the time-averaged allocations over the finite window size converge to a unique invariant measure. We also show that the ergodic property holds for the model.**

***Keywords*— Distributed optimization, AIMD, multi-resource allocation, Ergodicity, Invariant measure.**

I. Introduction

Resource allocation is a fundamental problem that arises in many application areas, for example, smart energy systems, edge computing, communication networks, etc. In many cases, agents may have limited resources and may require multiple shared resources to complete a task. Additionally, the agents may have heterogeneous resource demands and aim to maximize network utility. For such distributed multi-resource allocation problems, single distributed resource allocation solutions provide sub-optimal solutions [1]. Especially when the cost function of an agent is coupled through the allocation of multiple resources and the cost functions are non-separable—the functions that can not be separated into independent functions. It becomes more challenging when the agents demand resources based on their own preferences and local computation. Additionally, they do not want to exchange information with other agents in the network but with a central server that aggregates the demands from agents in the network and sends feedback signals if resource capacity is reached. Such setting is called *federated setting*; it has attracted significant interest from the research community [2]–[4].

In this paper, we formulate multi-resource allocation problems as distributed optimization problems. Moreover, we present distributed iterative multi-resource allocation algorithms wherein several agents in a network collaborate to access shared resources and minimize the total cost over the network. Roughly speaking, in the existing literature on distributed optimization, the agents achieve optimal allocation of resources through regular communication by exchanging their gradients of the cost functions [5]–[10] or sharing the Lagrange multipliers [11]–[13] with at least one of their neighbors, which may lead the agents to reveal their private information. Moreover, sharing information may also increase the communication overhead on the system.

We develop the additive increase and multiplicative decrease (AIMD)-based distributed iterative solutions to address these issues. Chiu and Jain [14] first proposed the AIMD algorithm for congestion control in TCP. In the AIMD algorithms, agents demand the resources linearly until they receive a *capacity event notification* from a central server; this is called the additive increase (AI) phase. The central server sends a *capacity event notification* when the aggregate demands of a resource exceed the resource capacity. After receiving the capacity event notification, agents in the network decrease their demands probabilistically; this is called the multiplicative decrease (MD) phase. After the MD phase, the agents again increase their demands until they receive the next capacity event notification. This process repeats [15], [16]. This paper extends these results and models the system as a homogeneous Markov chain with place-dependent probabilities over a fixed window size to show the convergence to a unique invariant measure. Now for $i = 1, 2, \ldots, n$, let agent i have a cost function $f_i : \mathbb{R}_+^m \to \mathbb{R}_+$ that is coupled through the allocation of m shared resources. Moreover, the multi-resource allocation problem can be formulated as the following optimization problem.

Problem 1 (Optimization problem)**.**

$$\begin{aligned} \underset{y_{11},\ldots,y_{mn}}{\text{minimize}} \quad & \sum_{i=1}^{n} f_i(y_{1i}, y_{2i}, \ldots, y_{mi}), \\ \text{subject to} \quad & \sum_{i=1}^{n} y_{ji} = \mathcal{C}_j, \quad \textit{for } j = 1, 2, \ldots, m, \\ & y_{ji} \geq 0, \quad \textit{for } i = 1, \ldots, n, \textit{ and } j = 1, \ldots, m. \end{aligned}$$

Our contribution to this paper is to formulate the multi-resource allocation problem as a distributed optimization problem. Furthermore, we model the system as a homogeneous Markov chain with place-dependent probabilities for the average allocation over a fixed-window size. Given the basic settings and assumptions on the probability distribution with which an agent responds to a capacity event of a resource, we show that the average allocations over the fixed-window size converge to a *unique invariant measure*. Furthermore, we show that the *ergodic property* holds. Note that this work and our work in [17] and [18] show convergence to a unique invariant measure and also show that ergodic property holds for average allocations over a fixed size window. In [18], we considered that the capacity events of resources occur in sequence, and we modeled the system as a homogeneous Markov chain. We later realized that applying Barnsley's contraction theorem to prove the convergence to a unique invariant measure and ergodic behavior with place-dependent response probabilities on this model is not a correct choice. However, the model works with fixed probabilities. Furthermore, the major difference between this work and our work in [17] is the algorithm and the way the system is modeled. In [17], we consider that after the occurrence of $M \in \mathbb{N}$ fixed capacity events of one resource, the agents keep their demands the same for that resource until M capacity events of another resource occur; then, all agents respond synchronously, we call these events *mega events*. This process repeats. However, the current paper considers the natural occurrence of capacity events at time instants, and they are not forced to occur. In the current approach, both resources may have capacity events at one time-instant. We model the system at these naturally occurring synchronous capacity events to show convergence.

II. Problem formulation

Suppose that n agents coordinate to access m limited divisible and heterogeneous resources. Let $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_m$ denote the capacities of resources $1, 2, \ldots, m$, respectively. Each agent associates a cost to a specific allocation of resources. Let $\mathbb{R}$ denote the set of real numbers and $\mathbb{R}_+$ denote the set of non-negative real numbers. Let $\mathbb{R}^n$ denote the column vectors of length n with real-valued entries. Moreover, the n-dimensional vector $\mathbf{e}_j = [0, \ldots, 1, \ldots, 0]^\top$ is a canonical basis vector with 1 at the j'th place. Let $t \in \mathbb{R}_+$ represent time. We denote the sets $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}_+ = \{1, 2, 3, \ldots\}$. Agents are indexed by $i = 1, 2, \ldots, n$, and resources are indexed by $j = 1, 2, \ldots, m$. Let $x_{ji}(t) \in [0, \mathcal{C}_j]$ denote the amount of resource j allocated to agent i at time t, for $i = 1, 2, \ldots, n$ and $j = 1, 2, \ldots, m$. We assume that the cost function f_i satisfies Assumption 1.

Assumption 1. *The cost function $f_i : \mathbb{R}_+^m \to \mathbb{R}_+$ is continuously differentiable, strictly convex, and increasing in each variable.*

Let $t_0 < t_1 < t_2 < \ldots$ be the discrete time steps. Let $k_j \in \mathbb{N}$ denote the capacity event of resource j, and t_{k_j} denote the time instant at which the k_j'th capacity event occurs. Let $W \in \mathbb{N}_+$ be a fixed time window (the number of capacity events of a resource) over which we want to model the system. Let $\overline{\mathbf{x}}_{jW}(t_{k_j}) \in \mathbb{R}^n$ be the time-averaged allocation of resource j over the time-window W. For $j = 1, 2, \ldots, m$, and $k_j \in \mathbb{N}$, we define $\overline{\mathbf{x}}_{jW}(t_{k_j})$ as follows

$$\overline{\mathbf{x}}_{jW}(t_{k_j}) \triangleq \frac{1}{W} \sum_{\ell=0}^{W-1} \mathbf{x}_j(t_{k_j} - t_\ell). \tag{1}$$

For $k \in \mathbb{N}$, let $\boldsymbol{\xi}(t_k)$ be defined as

$$\boldsymbol{\xi}(t_k) \triangleq [\overline{\mathbf{x}}_{1W}(t_k)^\top \quad \overline{\mathbf{x}}_{2W}(t_k)^\top \quad \ldots \quad \overline{\mathbf{x}}_{mW}(t_k)^\top]^\top. \tag{2}$$

Here, we consider the average allocations over W capacity events of resources until time instant t_k. We use simplex Σ throughout the paper, defined as:

Definition 1 (Simplex)**.** *Let Σ be the simplex in $\mathbb{R}^n$, for $x_i \geq 0, i = 1, 2, \ldots, n$, we define it as:*

$$\Sigma \triangleq \big\{\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}_+^n \;\big|\; \sum_{i=1}^n x_i = 1\big\}. \tag{3}$$

Analogously, we define simplex Σ_W in $(\mathbb{R}^n)^W$. We present the multi-variate AIMD matrix in Section III. Furthermore, we model the system as a homogeneous Markov chain with place-dependent probabilities; the response probabilities depend on the average allocation $\overline{\mathbf{x}}_{jW}(t_{k_j})$ for finite fixed-window size W. Now, let π^W be an *invariant probability distribution* on $\big(\Sigma_W\big)^m$, where m is the number of resources in the system. We aim to show that the proposed distributed and randomized algorithm achieves the following goal,

(i) for every finite window size $W \geq 1$, there exists a unique invariant measure π^W on $\big(\Sigma_W\big)^m$, and,
(ii) for every $\boldsymbol{\xi}(t_k) \in \big(\Sigma_W\big)^m$ and given initial value $\boldsymbol{\xi}(t_0) \in \big(\Sigma_W\big)^m$, the ergodic property holds, that is,

$$\lim_{k\to\infty} \frac{1}{k+1} \sum_{\ell=0}^{k} \boldsymbol{\xi}(t_\ell) = \mathbb{E}(\pi^W), \text{ almost surely,}$$

where $\mathbb{E}(\cdot)$ represents the expected value.

III. Multi-resource allocation algorithm

In this section, we briefly present the AIMD-based algorithm for allocating multiple divisible heterogeneous resources. The algorithm consists of two phases—*additive increase (AI)* and *multiplicative decrease (MD)*. In the AI phase, an agent increases its demand for a resource linearly by a positive constant called *additive increase factor* until the aggregate demand for the resource reaches the capacity of that resource. Afterward, a central server broadcasts a one-bit *capacity event signal* in the system. After receiving this signal, an agent responds in a probabilistic way, whether to decrease its demand for a resource by a constant called *multiplicative decrease factor* or not; this is called the MD phase. After the MD phase, again, the AI phase starts, and agents increase their resource demands linearly by additive

increase factors until they receive the next capacity event signal. This process repeats over time. More details can be found at [16] and [19].

Let $\alpha_j > 0$ be the additive increase factor and $0 \leq \beta_j < 1$ be the multiplicative decrease factor of an agent for resource j, considering all agents have the same multiplicative decrease factor, for $j = 1, 2, \ldots, m$. Furthermore, let $\boldsymbol{\beta}_{ji}$ be a random variable that denotes the multiplicative decrease factor of agent i for resource j. It takes value β_j, or 1, for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots, m$. For all j, let $\Gamma_j > 0$ be the *normalization factor* of resource j broadcasted at the start of the algorithm by the central server. After the start of the system, an agent increases its resource demand linearly by the constant α_j until the aggregate demand equals the resource capacity $\mathcal{C}_j$. Once the aggregate demand reaches the resource capacity, the central server broadcasts a *one-bit* feedback signal to notify the agents to reduce their demands for the resource j, for $j = 1, 2, \ldots, m$; we call this signal a *capacity event* signal. Hence, for $k_j \in \mathbb{N}_+$ and time $t \in (t_{(k-1)_j}, t_{k_j}]$, we formulate (the AI phase) as follows:

$$x_{ji}(t) = x_{ji}(t_{(k-1)_j}) + (t - t_{(k-1)_j})\alpha_j, \quad \text{for } i = 1, 2, \ldots, n, \text{ and } j = 1, 2, \ldots, m. \tag{4}$$

For the sake of simplicity of notation, at the k_j^{th} capacity event, we use $x_{ji}(k_j)$ to represent $x_{ji}(t_{k_j})$, and analogously for other variables, for all j and k_j. Now, let $\sigma_{ji}(k_j)$ (cf. (7)) denote the response probability with which agent i responds to the k_j'th capacity event then we formulate that $\mathbb{P}\big(\boldsymbol{\beta}_{ji}(k_j) = \beta_j\big) = \sigma_{ji}(k_j)$ and $\mathbb{P}\big(\boldsymbol{\beta}_{ji}(k_j) = 1\big) = 1 - \sigma_{ji}(k_j)$, for $i = 1, 2, \ldots, n$; $j = 1, 2, \ldots, m$; and $k_j \in \mathbb{N}$. After receiving the capacity event notification, agent i responds by multiplicatively decreasing its demand by $\boldsymbol{\beta}_{ji}(k_j) \in \{\beta_j, 1\}$ as

$$x_{ji}(t) = \boldsymbol{\beta}_{ji}(k_j) x_{ji}(t_{k_j}) \quad \text{if } t = t_{k_j}^+; \tag{5}$$

this is the MD phase. We now define the time-averaged allocation as

$$\overline{x}_{ji}(k_j) \triangleq \frac{1}{k_j + 1} \sum_{\ell=0}^{k_j} x_{ji}(\ell). \tag{6}$$

For all i, j, and k_j, the response probability $\sigma_{ji}(k_j)$ is obtained by

$$\sigma_{ji}(k_j) = \Gamma_j \frac{\frac{\partial}{\partial_x}\big|_{x=\overline{x}_{ji}(k_j)} f_i\big(\overline{x}_{1i}(k_j), \ldots, \overline{x}_{mi}(k_j)\big)}{\overline{x}_{ji}(k_j)}. \tag{7}$$

Note that for all j, the normalization factor Γ_j is calculated by the central server and sent by it to the agents when the agents join the network. Γ_j is used to keep the probability $\sigma_{ji}(k_j)$ in $(0, 1]$, for all i and k_j. After the MD phase, again agents enter into the AI phase until they receive the next capacity event notification. This process repeats.

IV. AIMD MATRIX FOR MULTIPLE RESOURCES

In this section, we present the AIMD matrix formulation for multiple resources. We use $(t_{(k+1)_j} - t_{k_j})$ to denote the time difference between the k_j'th and the $(k_j + 1)$'th capacity events, for $k_j \in \mathbb{N}$. We now derive the AIMD matrix for multiple variables and model the system as a homogeneous Markov chain with place-dependent probabilities. In (8), $x_{ji}(k_j + 1)$ represents the allocation at the $(k_j + 1)^{\text{th}}$ capacity event, and it is written as

$$x_{ji}(k_j + 1) = \boldsymbol{\beta}_{ji}(k_j) x_{ji}(k_j) + (t_{(k_j+1)_j} - t_{k_j})\alpha_j, \tag{8}$$

where $\mathbb{P}\big(\boldsymbol{\beta}_{ji}(k_j) = \beta_j\big) = \sigma_{ji}(k_j)$ and $\mathbb{P}\big(\boldsymbol{\beta}_{ji}(k_j) = 1\big) = 1 - \sigma_{ji}(k_j)$, for $i = 1, 2, \ldots, n$; $j = 1, 2, \ldots, m$; and $k_j \in \mathbb{N}$.

With little algebraic manipulation and using the fact that $\sum_{i=1}^n x_{ji}(t_{k_j+1}) = \sum_{i=1}^n x_{ji}(t_{k_j}) = \mathcal{C}_j$, we can find the value of $(t_{(k+1)_j} - t_{k_j})$. Note that the time between two capacity events is a random variable. Let $\mathbf{x}_j = [x_{j1} \; x_{j2} \; \ldots \; x_{jn}]^\top$, and let $\mathbf{e} = [1 \quad 1 \quad \ldots \quad 1]^\top \in \mathbb{R}^n$. Then after replacing the value of $(t_{(k+1)_j} - t_{k_j})$ in (8), we obtain

$$\mathbf{A}_j(k_j) = \text{diag}\,(\boldsymbol{\beta}_{j1}(k_j), \boldsymbol{\beta}_{j2}(k_j), \ldots, \boldsymbol{\beta}_{jn}(k_j)) + \frac{1}{n}\mathbf{e}\left[1 - \boldsymbol{\beta}_{j1}(k_j) \quad \ldots \quad 1 - \boldsymbol{\beta}_{jn}(k_j)\right], \tag{9}$$

where $\boldsymbol{\beta}_{ji}(k_j) \in \{\beta_j, 1\}$, for $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, m$, and $k_j \in \mathbb{N}$. Note that the AIMD matrix in (9) is a simplified version obtained using the same additive increase and multiplicative decrease factors for all agents in the network. Moreover, we obtain

$$\mathbf{x}_j(k_j + 1) = \mathbf{A}_j(k_j)\mathbf{x}_j(k_j), \tag{10}$$

for $j = 1, 2, \ldots, m$ and $k_j \in \mathbb{N}$, where $\mathbf{A}_j(k_j) \in \mathbb{R}^{n \times n}$ is the matrix for resource j. The matrix $\mathbf{A}_j(k_j)$ has randomized entries, and it is called multi-variate *AIMD matrix*. It is a non-negative column stochastic matrix. For a fixed j, let $\text{diag}\,(\boldsymbol{\beta}_{j1}, \boldsymbol{\beta}_{j2}, \ldots, \boldsymbol{\beta}_{jn})$ denote the diagonal matrix with $\boldsymbol{\beta}_{ji} \in \{\beta_j, 1\}$ as a diagonal, for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots, m$. For $j = 1, 2, \ldots, m$ and $k \in \mathbb{N}$ and $\boldsymbol{\beta}_{ji} \in \{\beta_j, 1\}$, we define matrix A_j as

$$A_j \triangleq \text{diag}\,(\boldsymbol{\beta}_{j1}, \ldots, \boldsymbol{\beta}_{jn}) + \frac{1}{n}\mathbf{e}(\mathbf{e}^\top - [\boldsymbol{\beta}_{j1} \ldots \boldsymbol{\beta}_{jn}]), \tag{11}$$

which is a non-negative column stochastic matrix. Furthermore, we define matrix A_j^* wherein every agent decreases its resource demand by β_j:

$$A_j^* \triangleq \text{diag}\,(\underbrace{\beta_j, \ldots, \beta_j}_{n}) + \frac{1}{n}\mathbf{e}(\mathbf{e}^\top - [\underbrace{\beta_j \ldots \beta_j}_{n}]). \tag{12}$$

For $j = 1, 2, \ldots, m$, let $\mathcal{A}_j$ denote the set of all possible AIMD matrices of resource j; the set $\mathcal{A}_j$ consists of 2^n AIMD matrices for n agents in the network. Recall that the response probability σ_{ji} is the probability with which agent i responds to the capacity event of resource j, for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots, m$. For $A_j \in \mathcal{A}_j$, we obtain the following probability:

$$\mathbb{P}\big(\mathbf{A}_j(k_j) = A_j\big) = \prod_{\boldsymbol{\beta}_{ji} = \beta_j} \sigma_{ji}(k_j) \prod_{\boldsymbol{\beta}_{ji} = 1} (1 - \sigma_{ji}(k_j)), \tag{13}$$

assuming that the probability $\sigma_{ji}(k_j)$ is independent, for $i = 1, 2, \ldots, n; j = 1, 2, \ldots, m;$ and $k_j \in \mathbb{N}$.

V. Main results

In this section, we study the resource allocation model under the assumption that the response probability to capacity events depends on the average allocations over a finite time window. The model is a homogeneous Markov chain with place-dependent probabilities. Under the assumption that the response probability functions are strictly increasing in each variable, bounded away from zero, and are Lipschitz continuous, we show that the model has a unique invariant measure. We also show that an ergodic property holds. Note that the system works for m resources, but we consider two resources at some places for clarity of presentation.

Let $\Delta_1(t_{k_1}) \in \{0,1\}$, where $\Delta_1(t_{k_1}) = 1$ denotes the occurrence of the capacity event of resource 1 at time instant t_{k_1}, and $\Delta_1(t_{k_1}) = 0$ denotes that the capacity event did not occur at time instant t_{k_1}. Analogously, let $\Delta_2(t_{k_2}) \in \{0,1\}$, and $\Delta_2(t_{k_2}) = 1$ denotes the occurrence of capacity event of resource 2 at time instant t_{k_2}. We formulate the evolution of allocations of two resources at time instant t_k as follows:

$$\mathbf{A}(k) \triangleq \begin{cases} \text{diag}\ (\mathbf{A}_1(k_1), I) & \text{if } \Delta_1(t_{k_1}) = 1, \\ \text{diag}\ (I, \mathbf{A}_2(k_2)) & \text{if } \Delta_2(t_{k_2}) = 1, \\ \text{diag}\ (\mathbf{A}_1(k_1), \mathbf{A}_2(k_2)) & \text{if } \Delta_1(t_{k_1}) = 1 \text{ and } \Delta_2(t_{k_2}) = 1 \\ & \text{at } t_{k_1} = t_{k_2}. \end{cases} \tag{14}$$

Note that $\mathbf{A}_1(k_1)$ in (14) denotes the AIMD matrix of resource 1 over resource 1's capacity events up to time instant t_k. Analogously, we state for $\mathbf{A}_2(k_2)$. For $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2) \in \Sigma^2$, we obtain:

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k). \tag{15}$$

Let $T_1, T_2, \ldots,$ be the time instants at which capacity events of both resources occur synchronously. We assume that synchronous events occur finitely many times. Let up-to time instant T_k, k_1 capacity events of resource 1 and k_2 capacity events of resource 2 occur. Furthermore, let between T_k and T_{k+1} time instants $r_{1,k}$ capacity events of resource 1 and $r_{2,k}$ capacity events of resource 2 occur. Then we formulate the evolution of resource allocations at synchronous capacity events as follows (see the definition of $\mathbf{A}(k)$ in (14)):

$$\begin{aligned} \begin{bmatrix} \mathbf{x}_1(T_{k+1}) \\ \mathbf{x}_2(T_{k+1}) \end{bmatrix} &= \text{diag}\ (\mathbf{A}_1(k_1 + r_{1,k}), \mathbf{A}_2(k_2 + r_{2,k})) \\ &\ldots \text{diag}\ (\mathbf{A}_1(k_1), I)\, \text{diag}\ (I, \mathbf{A}_2(k_2)) \\ &\times \begin{bmatrix} \mathbf{x}_1(T_k) \\ \mathbf{x}_2(T_k) \end{bmatrix}. \end{aligned} \tag{16}$$

We reformulate (16) to obtain:

$$\begin{aligned} &\begin{bmatrix} \mathbf{x}_1(T_{k+1}) \\ \mathbf{x}_2(T_{k+1}) \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{A}_1(k_1 + r_{1,k}) \ldots \mathbf{A}_1(k_1) & 0 \\ 0 & \mathbf{A}_2(k_2 + r_{2,k}) \ldots \mathbf{A}_2(k_2) \end{bmatrix} \\ &\times \begin{bmatrix} \mathbf{x}_1(T_k) \\ \mathbf{x}_2(T_k) \end{bmatrix}. \end{aligned} \tag{17}$$

Let $\mathbf{A}_1(k_1 + r_{1,k}) \ldots \mathbf{A}_1(k_1)$ be denoted by $\Pi_{A_1}(T_k)$; analogously, let $\mathbf{A}_2(k_2 + r_{2,k}) \ldots \mathbf{A}_2(k_2)$ be denoted by $\Pi_{A_2}(T_k)$. Then we have

$$\begin{bmatrix} \mathbf{x}_1(T_{k+1}) \\ \mathbf{x}_2(T_{k+1}) \end{bmatrix} = \begin{bmatrix} \Pi_{A_1}(T_k) & 0 \\ 0 & \Pi_{A_2}(T_k) \end{bmatrix} \begin{bmatrix} \mathbf{x}_1(T_k) \\ \mathbf{x}_2(T_k) \end{bmatrix}. \tag{18}$$

We define

$$\Pi(T_k) \triangleq \begin{bmatrix} \Pi_{A_1}(T_k) & 0 \\ 0 & \Pi_{A_2}(T_k) \end{bmatrix}. \tag{19}$$

Thus, we obtain

$$\mathbf{x}_1(T_{k+1}) = \Pi(T_k)\mathbf{x}_1(T_k). \tag{20}$$

To model the system's evolution, we consider the last $W \in \mathbb{N}$ capacity events until time t_k, and define average allocation vector $\mathbf{z}_j(T_k) \in (\mathbb{R}^n)^W$ over the interval $[k - W + 1, k]$, for $j = 1, 2, \ldots, m$:

$$\begin{aligned} &\mathbf{z}_j(T_k) \\ &\triangleq \Big[\mathbf{x}_j(T_k)^\top\ \frac{1}{2}\big(\mathbf{x}_j(T_k) + \mathbf{x}_j(T_{k-1})\big)^\top \ldots \\ &\frac{1}{W}\big(\mathbf{x}_j(T_k) + \ldots + \mathbf{x}_j(T_{k-W+1})\big)^\top\Big]^\top. \end{aligned} \tag{21}$$

We define the averages over fixed window size W as follows:

$$\mathbf{z}_{j,W}(T_k) = \frac{1}{W} \sum_{\ell=0}^{W-1} \mathbf{x}_j(T_{k-\ell}), \quad j = 1, 2, \ldots, m. \tag{22}$$

For a given matrix $A_j \in \mathbb{R}^{n \times n}$, we define the matrix $D_{A_j} \in \mathbb{R}^{Wn \times Wn}$ as follows:

$$D_{A_j} \triangleq \begin{bmatrix} \Pi_{A_j} & 0 & 0 & \ldots & & 0 \\ \frac{1}{2}(\Pi_{A_j} + I) & 0 & 0 & \ldots & & \vdots \\ \frac{1}{3}\Pi_{A_j} & \frac{2}{3}I & 0 & \ldots & & \\ \vdots & 0 & \ddots & \ddots & & \vdots \\ & \vdots & & \ddots & & \\ \frac{1}{W}\Pi_{A_j} & 0 & 0 & \ldots & \frac{W-1}{W}I & 0 \end{bmatrix}. \tag{23}$$

For $j = 1, 2, \ldots, m$, we define the set of D_{A_j} matrices as $\mathcal{D}_j$. Let A_j^* be the matrices where all agents respond to the capacity event by reducing their demands. Let $D_{A_j}^*$ (cf. (23)) denote the matrix block of A_j^*. We state the following results on non-expansive and contraction properties.

Lemma 1. *[15, Non-expansiveness and contraction properties] Let $A_j \in \mathcal{A}_j$ and $D_{A_j} \in \mathcal{D}_j$. Then:*

(i) *D_{A_j} is non-expansive on $(\mathbb{R}^n)^W$ in the sense that for all $\mathbf{z} \in (\mathbb{R}^n)^W$, we have $\|D_{A_j}\mathbf{z}\| \leq \|\mathbf{z}\|$.*

(ii) *Let the linear subspace be defined by*

$$\Psi_j \triangleq \left\{\mathbf{z} \in (\mathbb{R}^n)^W \mid \mathbf{e}^\top \mathbf{z}_\ell = 0, \text{ for } \ell = 1, 2, \ldots, W\right\};$$

it is invariant under all $D_{A_j} \in \mathcal{D}_j$.

(iii) $D^*_{A_j}$ *is contractive on* Ψ_j *in the sense that for all non-zero* $\mathbf{z} \in \Psi_j$, *we have* $\left\|D^*_{A_j}\mathbf{z}\right\| < \|\mathbf{z}\|$.

□

Let $A_{j,\ell} \in \mathcal{A}_j$ denote a particular (ℓ'th) AIMD matrix in $\mathcal{A}_j$, for $\ell = 1, \ldots, 2^n$. For $j = 1, 2, \ldots, m$; $\mathbf{z}_{j,W}(T_k)$ is the average over capacity events of resource j up to time instant T_k. Furthermore, let $p_{D_j} : \Sigma^m \to [0, 1]$ be probability distribution functions for choosing a matrix $D_{A_j,\ell}$ from $\mathcal{D}_j$, for $\ell = 1, \ldots, 2^n$. The evolution of $\mathbf{z}_j$ (as in (21)) is given by

$$\mathbf{z}_j(T_{k+1}) = \mathbf{D}_{A_j}(T_k)\mathbf{z}_j(T_k), \tag{24}$$

where $\mathbf{D}_{A_j}(k_j)$ is a random variable taking values in $\mathcal{D}_j$ with probability:

$$\mathbb{P}\left(\mathbf{D}_{A_j}(T_k) = D_{A_j,\ell}\right) = p_{D_j}\left(\mathbf{z}_{1,W}(T_k), \ldots, \mathbf{z}_{m,W}(T_k)\right). \tag{25}$$

For a time window $W \in \mathbb{N}$, let the state vector $\boldsymbol{\xi}(T_k)$ be defined as:

$$\boldsymbol{\xi}(T_k) \triangleq \left[\mathbf{z}_1(T_k)^\top \ \mathbf{z}_2(T_k)^\top \ \ldots \ \mathbf{z}_m(T_k)^\top\right]^\top. \tag{26}$$

For clarity, we consider two resources, that is, $m = 2$. Let matrices $\mathbf{U}(T_k) \in \mathbb{R}^{2Wn \times 2Wn}$ be defined as

$$\mathbf{U}(T_k) \triangleq \operatorname{diag}\left(\mathbf{D}_{A_1}(T_k), \mathbf{D}_{A_2}(T_k)\right). \tag{27}$$

Furthermore, let U^* denote $\operatorname{diag}\left(D_{A_1^*}, D_{A_2^*}\right)$. Recall that A_j^* denotes the AIMD matrix where all agents reduce their resource demands by β_j, $j = 1, 2$. Let the matrix set $\mathcal{U} = \{U^*, U_1, U_2, \ldots, U_{2^{n+1}-1}\}$. We obtain the following Markov chain:

$$\boldsymbol{\xi}(T_{k+1}) = \mathbf{U}(T_k)\boldsymbol{\xi}(T_k), \quad \text{for } k \in \mathbb{N}, \tag{28}$$

with place-dependent probabilities as

$$\mathbb{P}\left(\mathbf{U}(T_k) = U_\ell\right) = p_U\left(\mathbf{z}_{1,W}(T_k), \mathbf{z}_{2,W}(T_k)\right), U_\ell \in \mathcal{U}. \tag{29}$$

The probability $p_U(\cdot)$ depends on the average allocation of all resources over the interval $[k - W + 1, k]$. We have the following assumption on the probability distribution function σ_{ji}:

Assumption 2. (i) *For* $i = 1, \ldots, n$ *and* $j = 1, \ldots, m$, *the response probability distribution function* $\sigma_{ji} : \mathbb{R}^m_+ \to [0, 1]$ *is Lipschitz continuous.*

(ii) *For* $i = 1, \ldots, n$ *and* $j = 1, \ldots, m$, *there exists* $\sigma_{j,\min} > 0$ *such that* $\sigma_{ji} \geq \sigma_{j,\min}$.

We formulate Equation (25) in terms of $\sigma_{ji}(z_{1i,W}(T_k), \ldots, z_{mi,W}(T_k))$ that corresponds to agents responding to resource j's capacity events. Furthermore, based on Assumption 2, we obtain $p_U = p_{D_1}p_{D_2} > (\sigma_{1,\min})^n(\sigma_{2,\min})^n$. In the following subsection, we show the non-expansive and contraction properties of the product of $U \in \mathcal{U}$ matrices.

A. Non-expansive and contraction properties

Let $\Pi^*_{A_j}$ denote the product of a finite number of matrices from the set $\mathcal{A}_j$, which contains at least one A_j^*. Recall that A_j^* is a column stochastic matrix with strictly positive entries.

Proposition 1. *For a fixed* j,

(i) *for all* $\mathbf{z} \in \mathbb{R}^n$, *we have* $\left\|\Pi_{A_j}\mathbf{z}\right\|_1 \leq \|\mathbf{z}\|_1$.

(ii) *For all non-zero* $\mathbf{z} \in \mathbb{R}^n$, *we have* $\left\|\Pi^*_{A_j}\mathbf{z}\right\|_1 < \|\mathbf{z}\|_1$.

Lemma 2. *For a fixed* j, *let* $\Psi_j \in (\mathbb{R}^n)^W$ *be the invariant subspace under all matrices* D_{A_j}.

(i) *For all* $\mathbf{z} \in \Psi_j$, *we have* $\left\|D_{A_j}\mathbf{z}\right\| \leq \|\mathbf{z}\|$.

(ii) *For all non-zero* $\mathbf{z} \in \Psi_j$, *we have* $\left\|D_{A_j^*}\mathbf{z}\right\| < \|\mathbf{z}\|$.

Proof: It is presented in the Appendix.

Lemma 3. *Let* $\Psi \triangleq \{(\mathbf{z}_1, \mathbf{z}_2) \in (\mathbb{R}^n)^{2W} \mid \mathbf{e}^\top\mathbf{z}_{1,\ell} = \mathbf{e}^\top\mathbf{z}_{2,\ell} = 0, \text{ for } \ell = 1, \ldots, W\}$ *be the invariant subspace under all* $U \in \mathcal{U}$.

(i) *For all* $\mathbf{z} \in \Psi$, *we have* $\|U\mathbf{z}\| \leq \|\mathbf{z}\|$.

(ii) *For all non-zero* $\mathbf{z} \in \Psi$, *we have* $\|U^*\mathbf{z}\| < \|\mathbf{z}\|$.

□

Proof:

(i) As $\mathbf{z}_1, \mathbf{z}_2 \in (\mathbb{R}^n)^W$ and $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2) \in \Psi$, we obtain $\max\{\|\mathbf{z}_1\|, \|\mathbf{z}_2\|\} = \|\mathbf{z}\|$. Moreover, for a fixed j, from Lemma 2 (i), we have $\left\|D_{A_j}\mathbf{z}_j\right\| \leq \|\mathbf{z}_j\|$. Hence, we obtain

$$\begin{aligned}\|U\mathbf{z}\| &= \max\left\{\|D_{A_1}\mathbf{z}_1\|, \|D_{A_2}\mathbf{z}_2\|\right\} \\ &\leq \max\{\|\mathbf{z}_1\|, \|\mathbf{z}_2\|\} = \|\mathbf{z}\|.\end{aligned}$$

(ii) For a fixed j, from Lemma 2 (ii), we have $\left\|D_{A_j^*}\mathbf{z}_j\right\| < \|\mathbf{z}_j\|$. For all non-zero $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2) \in \Psi$, we obtain

$$\begin{aligned}\|U^*\mathbf{z}\| &= \max\left\{\left\|D_{A_1^*}\mathbf{z}_1\right\|, \left\|D_{A_2^*}\mathbf{z}_2\right\|\right\} \\ &< \max\left\{\|\mathbf{z}_1\|, \|\mathbf{z}_2\|\right\} = \|\mathbf{z}\|.\end{aligned}$$

We now show the main result of this paper—the existence of a unique invariant measure and that an ergodic property holds, using the result on average contraction by Barnsley [20].

B. Invariant measure and ergodic properties

Theorem 1 (Existence of a unique invariant measure [20])**.** *Suppose that Assumption 2 holds. Let* S *be the invariant subspace under all* $U \in \mathcal{U}$. *Additionally, let there exist* $c < 1$ *and* $\eta > 0$ *such that*

(i) *for all* $\mathbf{z}, \mathbf{w} \in \Sigma_W \times \Sigma_W$, *we have*

$$\sum_{U \in \mathcal{U}} p_U(\mathbf{z}) \frac{\|U(\mathbf{z} - \mathbf{w})\|}{\|\mathbf{z} - \mathbf{w}\|} < 1, \text{ and,} \tag{30}$$

(ii) *for all* $\mathbf{z}, \mathbf{w} \in \Sigma_W \times \Sigma_W$, *we have*

$$\sum_{U \in \mathcal{U}, \|U(\mathbf{z}-\mathbf{w})\| \leq c\|\mathbf{z}-\mathbf{w}\|} p_U(\mathbf{z})p_U(\mathbf{w}) \geq \eta^2. \tag{31}$$

Then, the model has a unique invariant measure, and the probability distribution is attractive. □

Theorem 2 (Ergodic property [21]). *Assume that the probabilities p_U satisfy* (30) *and* (31)*, and for time-window $W \in \mathbb{N}$, let π^W be the unique invariant measure on $\Sigma_W \times \Sigma_W$. Then, for any initial value $\boldsymbol{\xi}(T_0) \in \Sigma_W \times \Sigma_W$, the following (ergodic property) holds,*

$$\lim_{k\to\infty} \frac{1}{k+1} \sum_{\ell=0}^{k} \boldsymbol{\xi}(T_\ell) = \mathbb{E}[\pi^W], \text{ almost surely.} \tag{32}$$

□

Theorem 3. *Suppose that Assumption 2 holds. Then,*

(i) *for every time window $W \in \mathbb{N}$, there exists a unique invariant measure π^W on $\Sigma_W \times \Sigma_W$, and,*

(ii) *for every $\boldsymbol{\xi}(T_0) \in \Sigma_W \times \Sigma_W$, the ergodic property* (32) *holds.*

□

Proof:

(i) Recall that $U^* \in \mathcal{U}$ denotes the diagonal matrix $\operatorname{diag}(D_{A_1^*}, D_{A_2^*})$. For all i and j, as $\sigma_{ji} \geq \sigma_{j,\min} > 0$, thus, we obtain $p_{U^*}(\mathbf{z}) \geq (\sigma_{1,\min})^n (\sigma_{2,\min})^n > 0$, for all $\mathbf{z} \in \Sigma_W \times \Sigma_W$. Hence, the probability of matrix $U^* \in \mathcal{U}$ is strictly positive.

For all $\mathbf{z}$, $\mathbf{w} \in \Sigma_W \times \Sigma_W$, we have

$$\begin{aligned}&\sum_{U\in\mathcal{U}} p_U(\mathbf{z}) \|U(\mathbf{z}-\mathbf{w})\| \\ &= p_{U^*}(\mathbf{z}) \|U^*(\mathbf{z}-\mathbf{w})\| + \sum_{U\in\mathcal{U}-\{U^*\}} p_U(\mathbf{z}) \|U(\mathbf{z}-\mathbf{w})\|.\end{aligned} \tag{33}$$

Notice that $\mathbf{z}-\mathbf{w}$ is in the invariant subspace under all $U \in \mathcal{U}$. After applying Lemma 3, we reformulate (33) to obtain

$$\begin{aligned}&\sum_{U\in\mathcal{U}} p_U(\mathbf{z}) \|U(\mathbf{z}-\mathbf{w})\| \\ &< p_{U^*}(\mathbf{z}) \|\mathbf{z}-\mathbf{w}\| + \sum_{U\in\mathcal{U}-\{U^*\}} p_U(\mathbf{z}) \|\mathbf{z}-\mathbf{w}\|.\end{aligned} \tag{34}$$

Since $\sum_{U\in\mathcal{U}} p_U(\mathbf{z}) = 1$; therefore, $p_{U^*}(\mathbf{z}) = 1 - \sum_{U\in\mathcal{U}-\{U^*\}} p_U(\mathbf{z})$. With little algebraic manipulation, from (34), we obtain the contraction on average:

$$\begin{aligned}&\sum_{U\in\mathcal{U}} p_U(\mathbf{z}) \|U(\mathbf{z}-\mathbf{w})\| \\ &< \left(1 - \sum_{U\in\mathcal{U}-\{U^*\}} p_U(\mathbf{z})\right) \|\mathbf{z}-\mathbf{w}\| \\ &+ \sum_{U\in\mathcal{U}-\{U^*\}} p_U(\mathbf{z}) \|\mathbf{z}-\mathbf{w}\| = \|\mathbf{z}-\mathbf{w}\|, (cf.(30)).\end{aligned}$$

Let $\eta > 0$ denote $(\sigma_{1,\min})^n (\sigma_{2,\min})^n$. Now, we have $p_{U^*}(\mathbf{z}) \geq (\sigma_{1,\min})^n (\sigma_{2,\min})^n = \eta$, for all $\mathbf{z} \in \Sigma_W \times \Sigma_W$. For all $c < 1$, and for all $\mathbf{z}, \mathbf{w} \in \Sigma_W \times \Sigma_W$, we obtain

$$\sum_{U\in\mathcal{U}, \|U(\mathbf{z}-\mathbf{w})\| \leq c\|\mathbf{z}-\mathbf{w}\|} p_U(\mathbf{z}) p_U(\mathbf{w}) \geq \eta^2. \tag{35}$$

Thus, we obtain (31). As $(\sigma_{1,\min})^n (\sigma_{2,\min})^n = \eta > 0$, the sum in (35) is strictly positive.
All conditions of Theorem 1 are satisfied; hence, we conclude that the model has an invariant measure and the distribution is attractive. Because of the attractivity property, the invariant measure is unique.

(ii) As shown in (i), the model has the contraction on average and based on our assumption that the response probability functions are strictly positive and are Lipschitz continuous, all conditions of Theorem 2 is satisfied. Hence, we conclude that the ergodic property (32) holds.

VI. Conclusion

We presented an AIMD-based multi-resource allocation solution that does not require agent-to-agent communication; however, the agents coordinate with a central server that aggregates the resource demands and sends one-bit capacity events in the network. We modeled the system as a homogeneous Markov chain with place-dependent probabilities considering average allocations over a fixed window size. We proved that the average allocations over the fixed window size converge to a unique invariant measure. We also proved that the ergodic property holds for the model. Extending this work to show an almost sure convergence of average allocations to optimal values would be interesting.

Acknowledgement

Thanks to George Atia for suggesting the simultaneous occurrence of the capacity events. The suggestion made the model simpler and facilitated the use of Barnsley's contraction theorem. We acknowledge partial support from Mitacs (grant number IT24468).

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APPENDIX

Proof: [Proof of Lemma 2]

(i) We obtain

$$\begin{aligned}
&\left\|D_{A_j}\mathbf{z}\right\| = \\
&\left\|\begin{bmatrix} \Pi_{A_j} & 0 & 0 & \dots & & 0 \\ \frac{1}{2}\left(\Pi_{A_j}+I\right) & 0 & 0 & \dots & & \vdots \\ \frac{1}{3}\Pi_{A_j} & \frac{2}{3}I & 0 & \dots & & \\ \vdots & 0 & \ddots & \ddots & & \vdots \\ & \vdots & & \ddots & & \\ \frac{1}{W}\Pi_{A_j} & 0 & 0 & \dots & \frac{W-1}{W}I & 0 \end{bmatrix}\begin{bmatrix}\mathbf{z}_1\\ \mathbf{z}_2\\ \vdots\\ \mathbf{z}_W\end{bmatrix}\right\| \\
&= \left\|\begin{bmatrix} \Pi_{A_j}\mathbf{z}_1 \\ \frac{1}{2}\left(\Pi_{A_j}+I\right)\mathbf{z}_1 \\ \frac{1}{3}\Pi_{A_j}\mathbf{z}_1+\frac{2}{3}\mathbf{z}_2 \\ \vdots \\ \frac{1}{W}\Pi_{A_j}\mathbf{z}_1+\frac{W-1}{W}\mathbf{z}_{W-1}\end{bmatrix}\right\| \\
&= \max\left\{\left\|\Pi_{A_j}\mathbf{z}_1\right\|_1, \left\|\frac{1}{2}\left(\Pi_{A_j}\mathbf{z}_1+\mathbf{z}_1\right)\right\|_1, \dots, \right. \\
&\left.\left\|\frac{1}{W}\Pi_{A_j}\mathbf{z}_1+\frac{W-1}{W}\mathbf{z}_{W-1}\right\|_1\right\} \\
&\leq \max\left\{\left\|\Pi_{A_j}\mathbf{z}_1\right\|_1, \frac{1}{2}\left(\left\|\Pi_{A_j}\mathbf{z}_1\right\|_1+\|\mathbf{z}_1\|_1\right), \dots, \right. \\
&\left.\frac{1}{W}\left\|\Pi_{A_j}\mathbf{z}_1\right\|_1+\frac{W-1}{W}\|\mathbf{z}_{W-1}\|_1\right\}.
\end{aligned}$$

From Proposition 1, we have $\left\|\Pi_{A_j}\mathbf{z}_j\right\|_1 \leq \|\mathbf{z}_j\|_1$. Thus, we obtain

$$\begin{aligned}
&\left\|D_{A_j}\mathbf{z}\right\| \\
&\leq \max\left\{\|\mathbf{z}_1\|_1, \frac{1}{2}\left(\|\mathbf{z}_1\|_1+\|\mathbf{z}_1\|_1\right), \dots, \right. \\
&\left.\frac{1}{W}\|\mathbf{z}_1\|_1+\frac{W-1}{W}\|\mathbf{z}_{W-1}\|_1\right\} \\
&= \max\left\{\|\mathbf{z}_1\|_1, \|\mathbf{z}_1\|_1, \dots, \frac{1}{W}\|\mathbf{z}_1\|_1 \right. \\
&\left.+\frac{W-1}{W}\|\mathbf{z}_{W-1}\|_1\right\}.
\end{aligned}$$

Without loss of generality, let us suppose that for a fixed $\nu \in \mathbb{N}$, $\|\mathbf{z}_\nu\|_1$ is the maximum value among all $\|\mathbf{z}_\ell\|_1$, $\ell = 1, 2, \dots, W$; thus, we obtain

$$\left\|D_{A_j}\mathbf{z}\right\| \leq \|\mathbf{z}_\nu\|_1 = \|\mathbf{z}\|.$$

(ii) For $\mathbf{z}_u \in \mathbb{R}^n_+$, following the steps as in (i) and applying $\left\|\Pi^*_{A_j}\mathbf{z}_u\right\|_1 < \|\mathbf{z}_u\|_1$ (see Proposition 1), for all non-zero $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_W) \in \Psi_j$, we obtain $\left\|D_{A^*_j}\mathbf{z}\right\| < \|\mathbf{z}\|$.